

3.5. Вычислить определенный интеграл $F = \int_{x_0}^{x_1} y dx$, методами прямоугольников, трапеций, Симпсона с шагами h_1, h_2 . Оценить погрешность вычислений, используя Метод Рунге-Ромберга:

$$1. \quad y = \frac{x}{2x+5}, \quad X_0 = -1, \quad X_k = 1, \quad h_1 = 0.5, \quad h_2 = 0.25;$$

$$2. \quad y = \frac{x}{(3x+4)^2}, \quad X_0 = 0, \quad X_k = 4, \quad h_1 = 1.0, \quad h_2 = 0.5;$$

$$3. \quad y = \frac{x}{(3x+4)^3}, \quad X_0 = -1, \quad X_k = 1, \quad h_1 = 0.5, \quad h_2 = 0.25;$$

$$4. \quad y = \frac{3x+4}{2x+7}, \quad X_0 = -2, \quad X_k = 2, \quad h_1 = 1.0, \quad h_2 = 0.5;$$

$$5. \quad y = \frac{1}{(2x+7)(3x+4)}, \quad X_0 = -1, \quad X_k = 1, \quad h_1 = 0.5, \quad h_2 = 0.25;$$

$$6. \quad y = \frac{x}{(2x+7)(3x+4)}, \quad X_0 = -1, \quad X_k = 1, \quad h_1 = 0.5, \quad h_2 = 0.25;$$

$$7. \quad y = \frac{1}{3x^2+4x+2}, \quad X_0 = -2, \quad X_k = 2, \quad h_1 = 1.0, \quad h_2 = 0.5;$$

$$8. \quad y = \frac{1}{x^2+4}, \quad X_0 = -2, \quad X_k = 2, \quad h_1 = 1.0, \quad h_2 = 0.5;$$

$$9. \quad y = \frac{x}{x^2+9}, \quad X_0 = 0, \quad X_k = 2, \quad h_1 = 0.5, \quad h_2 = 0.25;$$

$$10. \quad y = \frac{x^2}{x^2+16}, \quad X_0 = 0, \quad X_k = 2, \quad h_1 = 0.5, \quad h_2 = 0.25;$$

$$11. \quad y = \frac{1}{x^3+64}, \quad X_0 = -2, \quad X_k = 2, \quad h_1 = 1.0, \quad h_2 = 0.5;$$

$$12. \quad y = \frac{x}{x^3+8}, \quad X_0 = -1, \quad X_k = 1, \quad h_1 = 0.5, \quad h_2 = 0.25;$$

$$13. \quad y = \frac{x^2}{x^3-27}, \quad X_0 = -2, \quad X_k = 2, \quad h_1 = 1.0, \quad h_2 = 0.5;$$

$$14. \quad y = \frac{1}{x^4+16}, \quad X_0 = 0, \quad X_k = 2, \quad h_1 = 0.5, \quad h_2 = 0.25;$$

$$15. \quad y = \frac{x}{x^4+81}, \quad X_0 = 0, \quad X_k = 2, \quad h_1 = 0.5, \quad h_2 = 0.25;$$

$$16. \quad y = \frac{x^2}{x^4+256}, \quad X_0 = 0, \quad X_k = 2, \quad h_1 = 0.5, \quad h_2 = 0.25;$$

$$17. \quad y = \frac{1}{256-x^4}, \quad X_0 = -2, \quad X_k = 2, \quad h_1 = 1.0, \quad h_2 = 0.5;$$

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| 18. | $y = \frac{x}{16 - x^4},$ | $X_0 = -1, \quad X_k = 1, \quad h_1 = 0.5, \quad h_2 = 0.25;$ |
| 19. | $y = \frac{x^2}{625 - x^4},$ | $X_0 = 0, \quad X_k = 4, \quad h_1 = 1.0, \quad h_2 = 0.5;$ |
| 20. | $y = \frac{\sqrt{x}}{4 + 3x},$ | $X_0 = 1, \quad X_k = 5, \quad h_1 = 1.0, \quad h_2 = 0.5;$ |
| 21. | $y = \frac{\sqrt{x}}{(1 + 2x)^2},$ | $X_0 = 1, \quad X_k = 5, \quad h_1 = 1.0, \quad h_2 = 0.5;$ |
| 22. | $y = x\sqrt{2x + 3},$ | $X_0 = -1, \quad X_k = 1, \quad h_1 = 0.5, \quad h_2 = 0.25;$ |
| 23. | $y = \frac{1}{\sqrt{(2x + 7)(3x + 4)}},$ | $X_0 = 0, \quad X_k = 4, \quad h_1 = 1.0, \quad h_2 = 0.5;$ |
| 24. | $y = \sqrt{16 - x^2},$ | $X_0 = -2, \quad X_k = 2, \quad h_1 = 1.0, \quad h_2 = 0.5;$ |
| 25. | $y = x\sqrt{49 - x^2},$ | $X_0 = -2, \quad X_k = 2, \quad h_1 = 1.0, \quad h_2 = 0.5;$ |
| 26. | $y = x^2\sqrt{36 - x^2},$ | $X_0 = 1, \quad X_k = 5, \quad h_1 = 1.0, \quad h_2 = 0.5;$ |
| 27. | $y = \sqrt{9 + x^2},$ | $X_0 = 1, \quad X_k = 5, \quad h_1 = 1.0, \quad h_2 = 0.5;$ |
| 28. | $y = x^3\sqrt{4 + x^2},$ | $X_0 = 1, \quad X_k = 5, \quad h_1 = 1.0, \quad h_2 = 0.5;$ |
| 29. | $y = \sqrt{x^2 - 36},$ | $X_0 = 6.5, \quad X_k = 8.5, \quad h_1 = 0.5, \quad h_2 = 0.25;$ |
| 30. | $y = x^3\sqrt{x^2 - 49},$ | $X_0 = 7.5, \quad X_k = 9.5, \quad h_1 = 0.5, \quad h_2 = 0.25;$ |